SMA Slope and Intercept

Overview of `sma_slope()` and `sma_intercept()`

Pressure volume (PV) curve parameter estimation relies on the estimation of points on standard major axis (SMA) regressions for water content and pressure (i.e., solute, pressure and leaf water potential). The SMA regression assumes that the primary error in estimating parameters in the y variable. In most cases with respect to the PV curve, this represent the pressure variable (Warton et al. 2006). The functions described here are used in the estimation of the saturated water content (SWC), osmotic potential at full turgor (π_o), apoplastic fraction (af), relative water content at turgor loss point (RWC_{tlp}) and leaf water potential at turgor loss point (Ψ_{tlp}). Capacitance at full turgor and at turgor loss also use these functions only represent the relationship between pressure potential and relative water content. These functions are used primarily as utility functions to keep the rest of the functions clean. Throughout this package, the x and y variables are changed depending on the variable that is being estimated.

Estimate Standard Major Axis slope (`sma_slope`)

The slope of the relationship is determined as the the standard deviation of the x variable divided by the standard deviation of the y variable (Sack and Pasquet-kok 2011):

$$\alpha = \frac{\sigma(x)}{\sigma(y)}$$

This is always the first parameter of the linear relationship to be estimated because it is used to estimate the intercept.

Usage

```
sma_slope(x,y)
```

- **x** *x variable*
- **y** *y variable*

Example

```
set.seed(1)# set seed for reproducibility

#create x and y variable
x<-rnorm(10)
y<-rnorm(10)

#estimate SMA slope
slope=pvest::sma_slope(x=x,
```

```
                    y=y)

#print slope
slope
#> [1] 0.7298505
```

Estimation of Standard Major Axis Intercept (`sma_intercept`)

With the slope estimated, we can then estimate the intercept of the relationship. The intercept of the SMA relationship is estimated as the mean of the x variable minus the estimated SMA slope multiplied by the mean value of the y variable.

$$b = mean(x) - \alpha * mean(y)$$

Usage

```
sma_intercept(x, y, slope)
```

- **x** *x variable*
- **y** *y variable*
- **slope** *SMA slope*

Example

```
#estimate SMA intercept
intercept<-pvest::sma_intercept(x=x,
                      y=y,
                      slope=slope)
#print intercept
intercept
#> [1] -0.04941685
```

References

Sack, Lawren, and Jessica Pasquet-kok. 2011. “Leaf Pressure-Volume Curve Parameters.” *PrometheusWiki*. http://prometheuswiki.org/tiki-pagehistory.php?page=Leaf%20pressure-volume%20curve%20parameters&preview=16.

Warton, David I., Ian J. Wright, Daniel S. Falster, and Mark Westoby. 2006. “Bivariate Line-Fitting Methods for Allometry.” *Biological Reviews* 81 (2): 259–91. https://doi.org/10.1017/S1464793106007007 10.1017/S1464793106007007</p>.

Estimation of Saturated and Relative Water Content

Overview of `SaturatedWaterContent()` and `RelativeWaterCD()`

This is a discussion of `SaturatedWaterContent()` and `RelativeWaterCD()` , which estimate the saturated water content (SWC), relative water deficit (RWD) and relative water content (RWC) respective for all values of water mass through dehydration for a given leaf (terms, units and definitions are tabled in Table 1). These functions take user measured leaf water mass and leaf water potential(Ψ_{leaf}) and returns a data frame with the leaf specific SWC and RWCs.

Table 1. Terms, units and definitions

Term (Abbreviation)	Units	Definition
Saturated water content(SWC)	g	Mass of leaf water at full saturation. Estimated as the y intercept of leaf water potential~leaf water.
Relative water deficit (RWD)	$g*g^{-1}$	Mass of water loss relative total saturation.
Relative water content (RWC)	$g*g^{-1}$	Mass of water relative to full saturation.
Leaf water potential (Ψ_{leaf})	MPa	Measured parameter which is an estimate of potential energy of water relative to pure water.
Leaf water mass(LWM)	g	Measured parameter estimated at the fresh mass of leaf minus dry mass and any other material weighed with leaf (e.g., whirlpak bag).

Saturated water content estimation(`SaturatedWaterContent()`)

The first step in performing a pressure volume curve analysis is estimating the saturated water content. It is used to estimate the relative water deficit and subsequently, the relative water content. To estimate SWC, we use the intercept of the SMA relationship (see `sma_slope()` and `sma_intercept()`) of leaf water and Ψ_{leaf} for the first four hydration states excluding any “plateau points”. Plateau points represent measurements of oversaturation of water within the leaf. It is evident when there is minimal change in leaf water potential with decreasing water content (LWM) (Kubiske and Abrams 1990). The relationship is represented as:

$$\Psi_{leaf} = \alpha * LWM + b,$$

where α and b represent the slope and intercept of the relationship:

$$\alpha = \frac{\sigma(LWM)}{\sigma(\Psi_{leaf})}$$

and,

$$b = mean(LWM) - \alpha * mean(\Psi_{leaf})$$

Usage

```
SaturatedWaterContent(data, fw.index, wp.index, dm.index)
```

- **data** a data frame containing leaf water potential and leaf water mass
- **fw.index** a numeric value indicating the column number where the leaf water mass data is within data frame
- **wp.index** a numeric value indicating the column number where the leaf water potential data is within data frame
- **dm.index** a numeric value indicating the column number where the leaf dry mass data is within data frame
- **NOTE:** Another important parameter is `n_row` . The default is `n_row=4` , meaning that the saturated water content will be estimated based on the first four hydration states (paired LWM and Ψ_{leaf}).

Example

```
#load data for Quercus agrifolia
quag1<-pvest::data%>%
  filter(leaf==1)

head(quag1)
#> # A tibble: 6 × 5
#>   species  leaf dry.weight water.potential fresh.weight
#>   <chr>   <dbl>      <dbl>           <dbl>        <dbl>
#> 1 quag_gj     1      0.201          -0.652        0.187
#> 2 quag_gj     1      0.201          -1.10         0.177
#> 3 quag_gj     1      0.201          -1.96         0.167
#> 4 quag_gj     1      0.201          -2.65         0.152
#> 5 quag_gj     1      0.201          -2.79         0.144
#> 6 quag_gj     1      0.201          -2.86         0.134

swm_swc<- SaturatedWaterContent(data=quag1,
                                fw.index = 5,
                                wp.index = 4,
                                dm.index = 3)
swm_swc # output list of saturated water mass and content
#> $SWM
#> [1] 0.1972973
#>
#> $SWC
#> [1] 0.9816761
```

Relative Water Content/Deficit Estimation (`RelativeWaterCD()`)

Once, the saturated water content is estimated, we can then express each leaf water mass relative to the mass of the leaf at full saturation(i.e., relative water content). RWC is determined for each dehydration state as:

$$RWC = \frac{LWM}{SWC}$$

For estimates of the components of leaf water potential, we also need the relative water deficit. We will also estimate that value for each RWC as:

$$RWD = 100 - RWC$$

Note that both RWC and RWD outputs from `RelativeWaterCD()` are expressed as percentages. `RelativeWaterCD()` takes as an input a data frame with a the saturated water content and leaf water mass and returns a data frame with RWC and RWD for further PV curve analysis.

Usage

```
RelativeWaterCD(data,fw.index)
```

- **data** a data frame containing leaf water potential and leaf water mass
- **fw.index** a numeric value indicating the column number where the leaf water mass data is within data frame

Example

```
#create column that contains the saturated water content and mass
quag1[,c("saturated.water.mass", "saturated.water.content")]<-swm_swc

#view full data frame with the new column
head(quag1)
#> # A tibble: 6 × 7
#>   species  leaf dry.weight water.potential fresh.weight saturated.water.mass
#>   <chr>   <dbl>      <dbl>           <dbl>        <dbl>                <dbl>
#> 1 quag_gj     1      0.201          -0.652        0.187                0.197
#> 2 quag_gj     1      0.201          -1.10         0.177                0.197
#> 3 quag_gj     1      0.201          -1.96         0.167                0.197
#> 4 quag_gj     1      0.201          -2.65         0.152                0.197
#> 5 quag_gj     1      0.201          -2.79         0.144                0.197
#> 6 quag_gj     1      0.201          -2.86         0.134                0.197
#> # ℹ 1 more variable: saturated.water.content <dbl>

#estimate RWC and RWD and add to data frame
quag1[,c("relative.water.content", "relative.water.deficit")]<-RelativeWaterCD(data=quag1,
                           fw.index = 5)

quag1[,c("relative.water.content", "relative.water.deficit")]
#> # A tibble: 9 × 2
#>   relative.water.content relative.water.deficit
#>                    <dbl>                  <dbl>
```

```
#> 1                    94.7                    5.29
#> 2                    89.6                   10.4
#> 3                    84.6                   15.4
#> 4                    76.8                   23.2
#> 5                    72.9                   27.1
#> 6                    68.0                   32.0
#> 7                    57.6                   42.4
#> 8                    53.1                   46.9
#> 9                    45.3                   54.7
```

References

Kubiske, M. E., and M. D. Abrams. 1990. “Pressure-Volume Relationships in Non-Rehydrated Tissue at Various Water Deficits.” *Plant, Cell & Environment* 13 (9): 995–1000. https://doi.org/10.1111/j.1365-3040.1990.tb01992.x 10.1111/j.1365-3040.1990.tb01992.x</p>.

Estimation of Osmotic Variables

The estimation of the osmotic variables (variables, units and symbols are presented in Table 1) is the second step **after** estimating the saturated water content and relative water content. These two parameters, especially relative water content, are used to estimate the osmotic variables using the linear relationship $\Psi_{leaf} = f(RWD)$. See `RelativeWaterCD` for information on the estimation of those parameters. The functions are an adaptation of the worksheet template from (Sack and Pasquet-kok 2011) and relationships drawn in (Bartlett, Scoffoni, and Sack 2012).

Table 1. Variables, units and symbols of osmotic estimates.

Symbol	Units	Definition
π_o	MPa	Osmotic potential at full turgor.
Max Ψ_p	MPa	Maximum pressure potential
Ψ_s	MPa	Osmotic potential for each hydration state
Ψ_p	MPa	Pressure potential
Ψ_{leaf}	MPa	Leaf water potential. Sum of leaf osmotic and pressure potential
a_f	%	Percent of total leaf water in the outside of cytoplasm
RWC_{sym}	$g*g^{-1}$	Mass of water relative to full saturation within the symplast. Estimated with respect to apoplastic fraction.

Estimate the π_o and regression parameters with `OsmoticPotFullTurgor()`

`OsmoticPotFullTurgor()` estimates the osmotic potential at full turgor from the linear relationship between relative water deficit and inverse leaf water potential. This function itself is only used within the `OsmoticEstimates()` function. This function estimates standard major axis(SMA) regression parameters from the values **below** turgor loss point. It assumes that your dataset is sufficient linear below turgor loss point to approximate osmotic potential at full turgor and a subsequent osmotic parameters. We assume that the data points at last four hydration states all represent points below turgor loss point and is the default.

The regression parameters are estimated by the helper functions(`sma_slope` and `sma_intercept`). The relationship is represented as:

$$\Psi_{leaf} = \alpha * RWD + b,$$

where α and b represent the slope and intercept of the relationship and RWD is the relative water deficit:

$$-\alpha = \frac{\sigma(RWD}{\sigma(\Psi_{leaf})}$$

and,

$$b = mean(RWD) - \alpha * mean(\Psi_{leaf})$$

$$RWD = 100 - Relative\ Water\ Content$$

Usage

```
OsmoticPotFullTurgor(data, fw.index, wp.index)
```

- **data** a data frame containing leaf water potential and leaf water mass
- **wc.index** a numeric value indicating the column number where the relative water content data is within data frame
- **wp.index** a numeric value indicating the column number where the leaf water potential(inverse) data is within data frame

Output

The function outputs a list of three numbers: slope and intercept of the line, and osmotic potential at full turgor.

Estimate osmotic parameters using `OsmoticEstimates()`

The `OsmoticEstimates()` function estimates the osmotic potential at full turgor (π_o) using `OsmoticPotFullTurgor()` and utilizes the slope and intercept values to estimate the other parameters (Table 1).

The function first selects the subset of the last for values for a given leaf then calculates the osmotic potential at full turgor and the maximum pressure potential as :

$$\pi_o = 1/intercept$$

$$max\ \Psi_p = \pi_o * -1$$

Because the leaf water potential at full hydration($\Psi_{leaf}=0$) is the sum of the osmotic potential at full turgor the maximum pressure potential, you can solve for the maximum pressure potential by solving for the Ψ_p when $\Psi_{leaf} = 0$.

The osmotic potential at a given hydration state is estimated by extrapolating the line of relative water deficit to inverse water potential:

$$\Psi_s = \frac{-1}{intercept + (slope * RWD)}$$

The pressure potential is the difference of the leaf water potential and the leaf osmotic potential:

$$\Psi_p = \Psi_{leaf} - \Psi_s$$

Other parameters estimated are the apoplastic fraction and the symplastic relative water content and deficit:

$$a_f = 100 + \frac{intercept}{slope}$$

$$RWC_{sym} = \frac{RWC - a_f}{100 - a_f} * 100$$

$$RWD_{sym} = 100 - RWC_{sym}$$

Usage

```
OsmoticEstimate(data, wc.index, wp.index)
```

- **data** a data frame containing leaf water potential and leaf water mass
- **wc.index** a numeric value indicating the column number where the relative water content data is within data frame
- **wp.index** a numeric value indicating the column number where the inverse leaf water potential data is within data frame

Example

```
#load data for Quercus agrifolia leaf 1 (QUAG 1)
quag1.df<-pvest::data%>%
  filter(leaf==1)%>%
  mutate(inv.water.potential=-1/water.potential)# calculate the inverse of water potential for RWC
         estimation

# estimate saturated water mass and saturated water content for QUAG 1
swc<- SaturatedWaterContent(data=quag1.df,
                                  fw.index = 5,
                                  wp.index = 4,
                                  dm.index = 3)

#create columns that contain the saturated water mass and saturated water content
quag1.df[,c("saturated.water.mass"," saturated.water.content")]<-swc

#estimate RWC and RWD
quag1.df[,c("relative.water.content", "relative.water.deficit")]<-RelativeWaterCD(data=quag1.df,
                                                                                  fw.index = 5)

q1_osm.est<-OsmoticEstimates(data=quag1.df, wc.index = "relative.water.deficit", wp.index =
         "inv.water.potential")

head(q1_osm.est%>%select(11:16))# show the estimated osmotic variables
#> # A tibble: 6 × 6
#>   osm.pot.fullturgor max.psip osmotic.potential pressure.potential
#>                <dbl>    <dbl>             <dbl>              <dbl>
#> 1              -2.08     2.08             -2.18             1.52
#> 2              -2.08     2.08             -2.28             1.18
#> 3              -2.08     2.08             -2.39             0.424
#> 4              -2.08     2.08             -2.58            -0.0677
#> 5              -2.08     2.08             -2.69            -0.0975
#> 6              -2.08     2.08             -2.85            -0.0122
#> # ℹ 2 more variables: apoplastic.fraction <dbl>, sym.rwc <dbl>
```

References

Bartlett, Megan K., Christine Scoffoni, and Lawren Sack. 2012. “The Determinants of Leaf Turgor Loss Point and Prediction of Drought Tolerance of Species and Biomes: A Global Meta-Analysis: Drivers of Plant Drought Tolerance.” *Ecology Letters* 15 (5): 393–405. https://doi.org/10.1111/j.1461-0248.2012.01751.x.

Sack, Lawren, and Jessica Pasquet-kok. 2011. “Leaf Pressure-Volume Curve Parameters.” *PrometheusWiki*. http://prometheuswiki.org/tiki-pagehistory.php?page=Leaf%20pressure-volume%20curve%20parameters&preview=16.

Estimation of the Turgor Loss Point

Estimate osmotic parameters at turgor loss

To estimate all the osmotic variables at turgor loss(i.e., relative water content/deficit at turgor loss and leaf water potential at turgor loss) for both the total and symplastic values, we start with predicting the line parameters for the Ψ_p versus relative water deficit. This process is similar to the previously used functions(see `vignette("sma-slope-intercept")` and `vignette("swc-and-rwc-estimation")`). We are using the the same SMA slope and intercept functions but changing the x and y inputs in the function (`help("PsiPRWD_slopeint")`).

Beyond that, we need to use the linear relationship below turgor loss so the function (`EstimateTLP()`) also sources `OsmoticPotFullTurgor()` to get the slope and intercept parameters. For more information on that that function see `help("OsmoticPotFullTurgor")` or `vignette('osmotic-estimates')`.

`EstimateTLP()` first slices the head and tail of the data by the number of rows(`n_row`). The data above turgor loss is utilized as the data for `PsiPRWD_slopeint()` and the data below turgor loss used to estimate the parameters from `OsmoticPotFullTurgor`. Then, the turgor loss parameters and modulus are estimated as:

$$RWD_{tlp} = -\frac{intercept}{slope}$$

$$RWD_{sym,tlp} = -\frac{intercept}{slope}$$

where the intercept and slope here are those estimated for the symplastic and total relative water deficit versus leaf pressure potential for the total and symplastic parameters, respectively. The respective relative water content values are estimated as $100 - RWD$. Leaf water potential at turgor loss point (Ψ_{tlp}) is estimate by solving for the water potential at the relative water deficit at turgor loss, for the total parameter:

$$\Psi_{tlp} = \frac{-1}{slope * RWD_{tlp} + intercept}$$

where the slope and intercept here are those of the relationship between inverse leaf water potential and relative water deficit. Lastly, the symplastic and total modulus of elasticity are the quotient of the maximum pressure potential and relative water content at turgor loss point:

$$\epsilon = \frac{\frac{max\Psi_p}{100-RWD_{tlp}}}{100}$$

$$\epsilon_{sym} = \frac{\frac{max\Psi_p}{100-RWD_{sym,tlp}}}{100}$$

Usage

```
EstimateTLP(data, wc.index, wp.index, n_row=4)
```

data a data frame containing leaf water potential and leaf water mass

wc.index a numeric value indicating the column number where the relative water content data is within data frame

wp.index a numeric value indicating the column number where the inverse leaf water potential data is within data frame

n_row a numeric value indicating the number of rows to filter above and below turgor loss

Example

```
#load data for Quercus agrifolia leaf 1 (QUAG 1)
quag1.df<-pvest::data%>%
  filter(leaf==1)%>%
  mutate(inv.water.potential=-1/water.potential)# calculate the inverse of water potential for RWC
         estimation

# estimate saturated water mass and saturated water content for QUAG 1
swc<- SaturatedWaterContent(data=quag1.df,
                                    fw.index = 5,
                                    wp.index = 4,
                                    dm.index = 3)

#create columns that contain the saturated water mass and saturated water content
quag1.df[,c("saturated.water.mass"," saturated.water.content")]<-swc

#estimate RWC and RWD
quag1.df[,c("relative.water.content", "relative.water.deficit")]<-RelativeWaterCD(data=quag1.df,
                                                                                  fw.index = 5)

q1_osm.est<-OsmoticEstimates(data=quag1.df, wc.index = "relative.water.deficit", wp.index =
         "inv.water.potential")

q1_tlp.est<-EstimateTLP(q1_osm.est, wc.index = "relative.water.deficit",wp.index =
         "inv.water.potential")

#summarize leaf level values
(q1_tlp.sum<-q1_tlp.est%>%
  select(1,2,10,11,14,17:23)%>%
  unique())
#> # A tibble: 9 × 12
#>   species  leaf relative.water.deficit osm.pot.fullturgor pressure.potential
#>   <chr>   <dbl>                  <dbl>              <dbl>              <dbl>
#> 1 quag_gj     1                   5.29              -2.08             1.52
#> 2 quag_gj     1                  10.4               -2.08             1.18
#> 3 quag_gj     1                  15.4               -2.08             0.424
#> 4 quag_gj     1                  23.2               -2.08            -0.0677
#> 5 quag_gj     1                  27.1               -2.08            -0.0975
#> 6 quag_gj     1                  32.0               -2.08            -0.0122
#> 7 quag_gj     1                  42.4               -2.08            -0.0634
#> 8 quag_gj     1                  46.9               -2.08             0.124
#> 9 quag_gj     1                  54.7               -2.08            -0.0528
#> # ℹ 7 more variables: sym.rwd <dbl>, relative.water.deficit.attlp <dbl>,
#> #   relative.water.content.attlp <dbl>, sym.rwd.attlp <dbl>,
#> #   sym.rwc.attlp <dbl>, leaf.waterpotential.attlp <dbl>, modulus <dbl>
```